CHEMISTRY-TT-02

Topics Covered: *Reversible and Irreversible Work; Heat Capacity; Adiabatic Processes and Poisson's Ratio*

Total Count of Questions: *48*

Expected Time of Attempt by Students: *90 min*

1. A 10 g piece of iron (C = 0.45 J/g°C) at 100°C is dropped into 25 g of water (C = 4.2 J/g°C at 27°C. Find temperature of the iron and water system at thermal equilibrium.

(a) 30°C (b) 33°C

(c) 40°C (d) None of these

2. At 25°C, a 0.01 mole sample of a gas is compressed from 4.0 L to 1.0 L at constant temperature. What is the work done for this process if the external pressure is 4.0 bar?

(a) 1.6×10^3 J (b) 8.0×10^2 J

(c) 4.0×10^2 J (d) 1.2×10^3 J

3. Calculate the work done (in J) when 4.5 g of H_2O_2 reacts against a pressure of 1.0 atm at 25°C

$2H_2O_2(l) \rightarrow O_2(g) + 2H_2O(l)$

(a) -1.63×10^2 (b) 4.5×10^2

(c) 3.2×10^2 (d) -6.1×10^2

4. 2 mole of zinc is dissolved in HCl at 25°C. The work done in open vessel is :

(a) –2.477 kJ (b) –4.955 kJ

(c) 0.0489 kJ (d) None

5. Temperature of 1 mole of a gas is increased by 2°C at constant pressure. Work done is :

(a) R (b) 2R

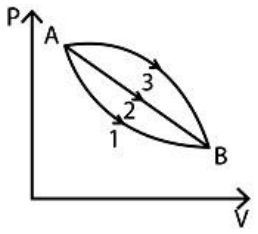
(c) R/2 (d) 3R

6. A sample of an ideal gas is expanded 1 m^3 to 3 m^3 in a reversible process for which $P = KV^2$, with K = 6 bar/m^6. Work done by the gas is:

(a) 5200 kJ (b) 15600 kJ

(c) 52 kJ (d) 5267.6 kJ

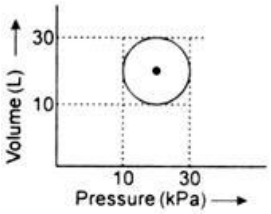
7. A given mass of gas expands reversibly from the state *A* to the state *B* by three paths 1, 2 and 3 as shown in the figure. If w_1, w_2 and w_3 respectively be the work done by the gas along three paths then:



(a) $w_1 > w_2 > w_3$ (b) $w_1 < w_2 < w_3$

(c) $w_1 = w_2 = w_3$ (d) $w_2 < w_3 < w_1$

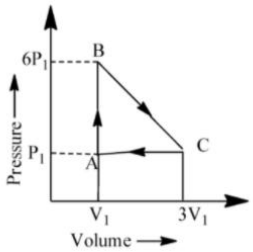
8. Heat absorbed by a system in going through a cyclic process shown in figure is :



(a) $10^7 \pi$ J (b) $10^6 \pi$ J

(c) $10^2 \pi$ J (d) $10^4 \pi$ J

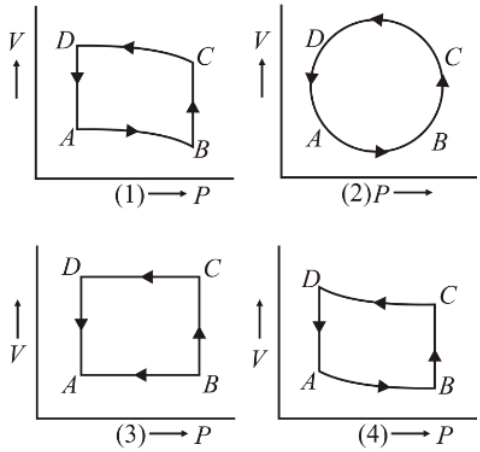
9. An ideal gas is taken around the cycle *ABCA* as shown in P–V diagram. The net work done during the cycle is equal to :



(a) $12P_1V_1$ (b) $6P_1V_1$

(c) $5P_1V_1$ (d) P_1V_1

10. In diagrams (1 to 4), variation of volume with changing pressure is shown. A gas is taken along the path *ABCD*. The change in internal energy of the gas will be:



(a) Positive in all the cases (1) to (4)
(b) Positive in cases (1), (2), (3) but zero in case (4)
(c) Negative in cases (1), (2), (3) but zero in case (4)
(d) Zero in all the cases

11. The temperature of an ideal gas increases in an :
(a) adiabatic expansion
(b) isothermal expansion
(c) adiabatic compression
(d) isothermal compression

12. For two mole of an ideal gas:
(a) $C_v - C_p = R$ (b) $C_p - C_v = 2R$
(c) $C_p - C_v = R$ (d) $C_v - C_p = 2R$

13. Which of the following expressions is true for an ideal gas?
(a) $\left(\frac{\partial V}{\partial T}\right)_P = 0$ (b) $\left(\frac{\partial P}{\partial T}\right)_V = 0$
(c) $\left(\frac{\partial U}{\partial V}\right)_T = 0$ (d) $\left(\frac{\partial U}{\partial T}\right)_V = 0$

14. A gas expands against a variable pressure given by $P = \frac{20}{V}$ (where P in atm and V in L). During expansion from volume of 1 litre to 10 litre, the gas undergoes a change in internal energy of 400 J. How much heat is absorbed by the gas during expansion?
(a) 46 J (b) 4660 J
(c) 5065.8 J (d) 4260 J

15. 2 mole of an ideal gas at 27°C expands isothermally and reversibly from a volume of 4 litre to 40 litre. The work done (in kJ) by the gas is:
(a) $w = -28.72$ kJ (b) $w = -11.488$ kJ
(c) $w = -5.736$ kJ (d) $w = -4.988$ kJ

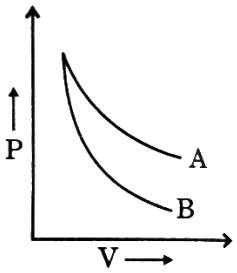
16. 10 mole of ideal gas expand isothermally and reversibly from a pressure of 10 atm to 1 atm at 300 K. What is the largest mass which can lifted through a height of 100 meter?
(a) 31842 kg (b) 58.55 kg
(c) 342.58 kg (d) None of these

17. The work done by the gas in reversible adiabatic expansion process is:
(a) $\frac{P_2V_2 - P_1V_1}{\gamma - 1}$ (b) $\frac{nR(T_1 - T_2)}{\gamma - 1}$
(c) $\frac{P_2V_2 - P_1V_1}{\gamma}$ (d) None of these

18. During an adiabatic process, the pressure of gas is found to be proportional to the cube of its absolute temperature. The ratio of $(C_{p,m}/C_{v,m})$ for gas is :
(a) $\frac{3}{2}$ (b) $\frac{5}{3}$
(c) $\frac{7}{2}$ (d) $\frac{4}{3}$

19. A gas expands adiabatically at constant pressure such that $T \propto V^{-1/2}$. The value of $\gamma(C_{p,m}/C_{v,m})$ of the gas will be:
(a) 1.30 (b) 1.50
(c) 1.70 (d) 2

20. P–V plot for two gases (assuming ideal) during adiabatic processes are given in the Fig. Plot A and plot B should correspond respectively to:



(a) He and H_2 (b) H_2 and He
(c) He and Ne (d) H_2 and Cl_2

21. Calculate the final temperature of a monoatomic ideal gas that is compressed reversible and adiabatically from 16 L to 2 L at 300 K:
(a) 600 K (b) 1044.6 K
(c) 1200 K (d) 2400 K

22. 5 mole of an ideal gas expand isothermally and irreversibly from a pressure of 10 atm to 1 atm against a constant external pressure of 1 atm. w_{irr} at 300 K is :
(a) –15.921 kJ (b) –11.224 kJ
(c) –110.83 kJ (d) None of these

23. The work done in adiabatic compression of 2 mole of an ideal monoatomic gas against constant external pressure of 2 atm starting from initial pressure of 1 atm and initial temperature of 300 K (R = 2 cal/mol-degree)
(a) 360 cal (b) 720 cal
(c) 800 cal (d) 1000 cal

24. One mole of an ideal gas $\left(C_{v,m}=\frac{5}{2}R\right)$ at 300 K and 5 atm is expanded adiabatically to a final pressure of 2 atm against a constant pressure of 2 atm. Final temperature of the gas is :

(a) 270 K (b) 273 K

(c) 248.5 K (d) 200 K

25. 10 litre of a non linear polyatomic ideal gas at 127°C and 2 atm pressure is suddenly released to 1 atm pressure and the gas expanded adiabatically against constant external pressure. The final temperature and volume of the gas respectively are.

(a) T = 350 K; V = 17.5 L

(b) T = 300 K; V = 15 L

(c) T = 250 K; V = 12.5 L

(d) None of these

26. Calculate average molar heat capacity at constant volume of gaseous mixture contained 2 mole of each of two ideal gases $A\left(C_{v,m}=\frac{3}{2}R\right)$ and $B\left(C_{v,m}=\frac{5}{2}R\right)$

(a) R (b) 2R

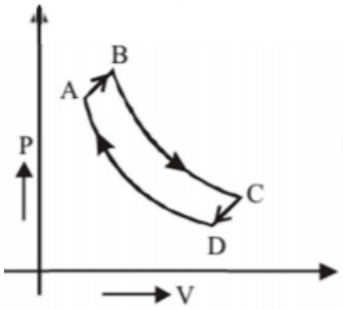
(c) 3R (d) 8R

27. 0.5 mole each of two ideal gases $A\left(C_{v,m}=\frac{5}{2}R\right)$ and $B\left(C_{v,m}=3R\right)$ are taken in a container and expanded reversibly and adiabatically, during this process temperature of gaseous mixture decreased from 350 K to 250 K. Find ΔH (in cal/mol) for the process :

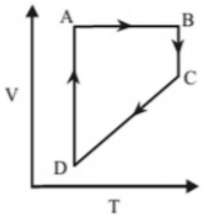
(a) –100 R (b) –137.5 R

(c) –375 R (d) None of these

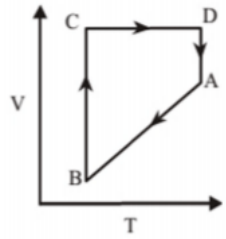
28. A cyclic process *ABCD* is shown in P–V diagram for an ideal gas. Which of the following diagram represents the same process?



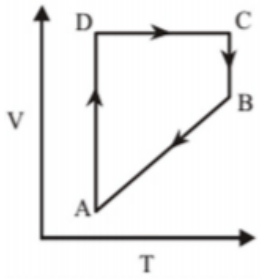
(a)



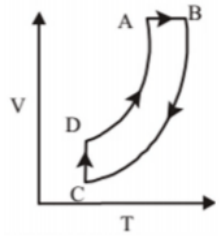
(b)



(c)



(d)



29. An adiabatic process is that process in which:

(a) energy is transferred as heat

(b) no energy is transfer red as heat

(c) ΔU = w

(d) the temp. of gas increases in a reversible adiabatic compression

30. In adiabatic process, the work involved during expansion or compression of an ideal gas is given by :

(a) $nC_v\Delta T$

(b) $\frac{nR}{\gamma-1}(T_2-T_1)$

(c) $-nRP_{ext}\left[\frac{T_2P_1-T_2P_2}{P_1P_2}\right]$

(d) $-2.303RT\log\frac{V_2}{V_1}$

31. If a process takes place in two steps at two different constant external pressures (which is less than internal pressure), then it can be called
(a) Two stages of irreversible process
(b) Single stage irreversible process
(c) Two stage reversible process
(d) Single stage reversible process

32. If a process is taking place at $P_{ex} = 0$, then it is a
(a) Free expansion
(b) Irreversible process
(c) Reversible process
(d) Both (a) and (b)

33. The internal energy change when a system goes from state A to B is 40 kJ/mole. If the system goes from A to B by a reversible path and returns to state A by an irreversible path what would be the net change in internal energy?
(a) > 40 kJ (b) < 40 kJ
(c) Zero (d) 40 kJ

34. A gas is allowed to expand in a well-insulated container against a constant external pressure of 2.5 atm from an initial volume of 2.50Lto a final volume of 4.50L. The change in internal energy ΔU of the gas in joules will be:-
(a) 500 J (b) -505 J
(c) + 505 J (d) 1136.25 J

35. When 1 mol gas is heated at constant volume, temperature is raised from 298 to 308 K. Heat supplied to the gas is 500 J. Then which statement is correct:
(a) q = W = 500 J, Δ U = 0
(b) q = ΔU = 500 J, W = 0
(c) q = W = 500 J, ΔU = 500
(d) Δ U = 0, q = W = –500 J

36. Zeroth law of thermodynamics gives the concept of
(a) Pressure (b) Volume
(c) Temperature (d) Mass

37. An ideal gas expands in volume from 1×10^{-4} m^3 to 1×10^{-3} m^3 at 300 K against a constant pressure 1×10^6 Nm^{-2}.The work done is
(a) –900 J (b) –900 kJ
(c) 270 kJ (d) 900 kJ

38. A system absorbs 10 kJ of heat and does 4 kJ of work. The internal energy of the system
(a) Decreases by 6 kJ
(b) Increases by 6 kJ
(c) Decreases by 14 kJ
(d) Increases by 14 kJ

39. In a process, 701 J of heat is absorbed by a system and 394 J of work is done by the system. What is the change in internal energy for the process?
(a) Increases by 307J
(b) Decreases by 307 J
(c) Increase by 400J
(d) Decreases by 400 J

40. A system is changed from state A to state B by one path and from B to A by another path. If ΔU_1 and ΔU_2 are the corresponding changes in internal energy, then
(a) $\Delta U_1 + \Delta U_2 = +ve$
(b) $\Delta U_1 + \Delta U_2 = -ve$
(c) $\Delta U_1 + \Delta U_2 = 0$
(d) None of these

41. Internal energy of a system includes all types of energy except
(a) Translation energy
(b) Bonding energy
(c) Rotational energy
(d) Gravitational potential energy

42. During compression of a spring the work done is 20 kJ and 5 kJ escaped to the surroundings as heat. The change in internal energy, ΔU (in kJ) is:
(a) –12 (b) –15
(c) 15 (d) 12

43. A system gives out 20 J of heat and also does 40 J of work. Calculate the internal energy change.
(a) 60 J (b) –60 J
(c) 20 J (d) –20 J

44. Which one of the following equations correctly represent the first law of thermodynamics for the given processes involving an ideal gas? (Assume non-expansion work is zero)
(a) Cyclic process: q = +w
(b) Adiabatic process: ΔU = –w
(c) Isochoric process: ΔU = –q
(d) Isothermal process; q = –w

45. If a gas absorbs 200 J of heat and expands by 500 cm^3 against a constant pressure of 2×10^5 Nm^{-2}, then the change in internal energy is
(a) –300 J (b) –100 J
(c) +100 J (d) +300 J

46. Which of the following is not correct about internal energy?
(a) Internal energy is a state function.
(b) Absolute value of internal energy can be calculated experimentally.
(c) The internal energy of a system is made up of a number of energies.
(d) All are correct.

47. When 2 mole of an ideal gas is heated at constant volume the temperature is raised from 300 K to 310 K. If heat supplied to the gas is x J, then which of the following statement is correct?

(a) $\Delta U = 0; q = w = -xJ$

(b) $q = w \neq xJ; \Delta U = 0$

(c) $q = \Delta U = xJ; w = 0$

(d) $\Delta U = 0; q = w = xJ$

48. A gas absorbs 200 J heat and undergoes simultaneous expansion against a constant external pressure of 105 Pa. The volume changes from 4L to 5L. The change in the internal energy is nearly

(a) 300 J (b) 100 J

(c) –300J (d) –100 J

ANSWERS

1. (a)	**2. (d)**	**3. (a)**	**4. (b)**	**5. (b)**	**6. (a)**	**7. (b)**	**8. (c)**	**9. (c)**	**10. (d)**
11. (c)	**12. (b)**	**13. (c)**	**14. (c)**	**15. (b)**	**16. (b)**	**17. (a)**	**18. (a)**	**19. (b)**	**20. (b)**
21. (c)	**22. (b)**	**23. (b)**	**24. (c)**	**25. (a)**	**26. (b)**	**27. (c)**	**28. (a)**	**29. (b,c,d)**	**30. (a,b,c)**
31. (a)	**32. (d)**	**33. (c)**	**34. (b)**	**35. (b)**	**36. (c)**	**37. (a)**	**38. (b)**	**39. (a)**	**40. (c)**
41. (d)	**42. (c)**	**43. (b)**	**44. (d)**	**45. (c)**	**46. (b)**	**47. (c)**	**48. (b)**		